

Figure 7.11: Tetanus – opisthotonos

inhibitory neurotransmitters (glycine and gamma – amino butyric acid) across the synaptic junction. The toxin acts presynaptically, the abolition of spinal inhibition causes uncontrolled spread of impulses in CNS (Central Nerves System). This results in muscle rigidity and spasms (due to the simultaneous contraction of agonists and antagonists, in the absence of reciprocal inhibition Figure 7.11).

7.7.4 Pathogenesis

Clostridium tetani is the causative organism of tetanus or lock jaw disease. pathogenesis of *Clostridium tetani* was discussed in detail in flowchart 7.4.

Source of infection – Soil, dust, faeces.

Route of entry – Through wound

Incubation period – 6–12 days

7.7.5 Clinical Feature

It includes, pain and tingling at the site of wound, Lock jaw ortrismus (It is reduced opening of the jaws), Risus sardonicus (mouth kept slightly open), Dysphasia (impairment of the ability to speak or to understand language) and acute asphyxia.

7.7.6 Laboratory Diagnosis

Specimens: Wound swab, exudates or tissue from wound.



Myth: Rust causes tetanus if introduced into a wound – for example, by stepping on a rusty nail.

Fact: Rusty nails are more likely to be contaminated with tetanus endospores because they have been exposed to soil and dust longer than new, rust free nails. Any object that causes a wound, rusty or not, can inoculate the tissue with the bacterial endospores of *C. tetani*. Rust itself neither causes tetanus nor makes it worse.

Microscopy: Gram staining shows Gram positive bacilli with drumstick appearance.

Culture: The clinical specimen is inoculated on blood agar and incubated at 37°C for 24–48 hours under anaerobic conditions. The colonies are confirmed by gram staining, where it shows gram positive bacilli with drumstick appearance.

7.7.7 Treatment

Tetanus patients are treated in special isolated units, to protect them from noise and light which may provoke convulsions. The spasm can be controlled by diazepam



Clostridium difficile is the causative agent of **antibiotic associated colitis.** It is an acute colitis with or without pseudo membrane formation. It is an important complication in patients on oral antibiotic therapy. Many antibiotics have been incriminated but lincomycin and clindamycin are particularly prone to cause pseudomembranous colitis.

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